

- Given the following data, what is the number-average molecular weight and the second virial coefficient? ($T = 25^\circ\text{C}$; density = 1.0 g/cm^3).

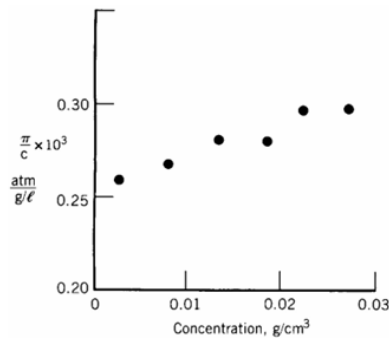


Figure P18.1 Osmotic pressure data.

- Estimate the Mark–Houwink–Sakurada constants from the data in Figure 3.15.

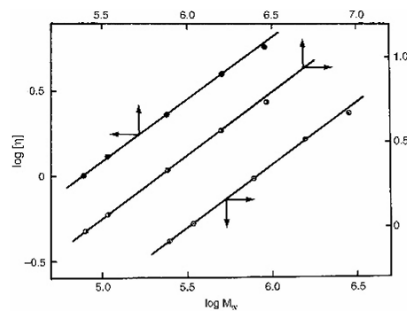


Figure 3.15 Double logarithmic plots of $[\eta]$ versus M_w for anionically synthesized polystyrenes, which were then fractionated leading to values of M_w/M_n of less than 1.06. Filled circles in benzene, half-filled circles in toluene, and open circles in dichloroethylene, all at 30°C (75). The arrows indicate the axes to be used. Units for $[\eta]$ in 100 ml/g .

- What is the glass transition temperature of a polymer, and how is it connected with molecular weights?
- Molecular weight estimated using Size-exclusion chromatography is an absolute method or a relative method?
- Why is vapour-pressure osmometry not a suitable method to estimate the ultra-high molecular weights?
- A polymer has the following molar mass. Calculate M_n , M_w , and polydispersity index.

No. of chains	Molecular weight (gm/mole)
40	5000
60	7000

- An equal number of protein mixtures containing Haemoglobin 15.5 Kg/mol , Ribonuclease 13.7 Kg/mol , Myoglobin 17.2 Kg/mol . Calculate the number average and mass average molecular weight of the protein solution.
- Intrinsic viscosity of polyisobutene is $180 \text{ cm}^3/\text{gm}$ and the Mark-Houwink constants K is 3.60×10^{-2} , α is 0.64. Estimate the molecular weight of polyisobutene.